

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Digital Signal and Image Processing Subject Code: 01CT1513	Aim :- Perform FFT and IFFT on discrete time signal.	
Lab-5	25-09-2026	Enrolment No:92400133131

Ami :-- Perform FFT and IFFT on discrete time signal.

Observation

1. The given discrete-time signal is converted from the **time domain to the frequency domain** using FFT.
2. FFT produces **complex-valued frequency components** containing real and imaginary parts.
3. The **magnitude spectrum** shows the strength of different frequency components present in the signal.
4. The **phase spectrum** represents the phase information of the frequency components.
5. The IFFT converts the frequency-domain FFT result back into the **time-domain signal**.
6. The reconstructed signal obtained using IFFT is approximately equal to the **original discrete-time signal**.
7. Therefore, FFT and IFFT successfully perform conversion between the **time domain and frequency domain** while preserving the signal information.

```
Code :- import numpy as np
import matplotlib.pyplot as plt
# =====
# INPUT SIGNAL
# =====
signal = np.array([1, 2, 3, 4, 5, 6, 7, 8])
N = len(signal)
n = np.arange(N)
# =====
# FFT
# =====
fft_result = np.fft.fft(signal)
# Real and imaginary parts
fft_real = np.real(fft_result)
fft_imag = np.imag(fft_result)
# Magnitude and phase
```

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```

magnitude_spectrum = np.abs(fft_result)
phase_spectrum = np.angle(fft_result)
# =====
# IFFT
# =====
reconstructed_signal = np.real(np.fft.ifft(fft_result))
# =====
# PRINT RESULTS
# =====
print("=" * 60)
print("                FFT ANALYSIS")
print("=" * 60)
print("\nOriginal Signal:")
print(signal)
print("\nFFT Result:")
for k, value in enumerate(fft_result):
    print(
        f"X[{k}] = "
        f"{value.real:.2f} "
        f"{'+' if value.imag >= 0 else '-'} "
        f"j{abs(value.imag):.2f}"
    )
print("\nMagnitude Spectrum:")
print(np.round(magnitude_spectrum, 2))
print("\nPhase Spectrum:")
print(np.round(phase_spectrum, 2))
print("\nReconstructed Signal:")
print(np.round(reconstructed_signal, 2))
# =====
# FIGURE 1 – ORIGINAL SIGNAL
# =====
plt.figure(figsize=(10, 5))
plt.stem(
    n,
    signal,
    linefmt='C0-',
    markerfmt='C0o',
    basefmt=' '
)

```

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for i, value in enumerate(signal):
    plt.annotate(
        f"{value}",
        (i, value),
        xytext=(0, 8),
        textcoords="offset points",
        ha="center",
        fontsize=10
    )
plt.title(
    "Original Discrete-Time Signal",
    fontsize=15,
    fontweight="bold"
)
plt.xlabel("Sample Index (n)")
plt.ylabel("Amplitude")
plt.xticks(n)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
# =====
# FIGURE 2 – FFT REAL PART
# =====
plt.figure(figsize=(10, 5))
plt.stem(
    n,
    fft_real,
    linefmt='C0-',
    markerfmt='C0o',
    basefmt=' '
)
for i, value in enumerate(fft_real):
    offset = 10 if value >= 0 else -18
    plt.annotate(
        f"{value:.2f}",
        (i, value),
        xytext=(0, offset),
        textcoords="offset points",
        ha="center",

```

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        fontsize=10
    )
plt.axhline(0, linewidth=0.8)
plt.title(
    "FFT - Real Part",
    fontsize=15,
    fontweight="bold"
)
plt.xlabel("Frequency Index (k)")
plt.ylabel("Real Part")
plt.xticks(n)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
# =====
# FIGURE 3 – FFT IMAGINARY PART
# =====
plt.figure(figsize=(10, 5))
plt.stem(
    n,
    fft_imag,
    linefmt='C1-',
    markerfmt='C1s',
    basefmt=' '
)
for i, value in enumerate(fft_imag):
    if value >= 0:
        offset = 10
    else:
        offset = -18
    plt.annotate(
        f"{value:.2f}j",
        (i, value),
        xytext=(0, offset),
        textcoords="offset points",
        ha="center",
        fontsize=10
    )
plt.axhline(0, linewidth=0.8)

```

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```
plt.title(
    "FFT - Imaginary Part",
    fontsize=15,
    fontweight="bold"
)
plt.xlabel("Frequency Index (k)")
plt.ylabel("Imaginary Part")
plt.xticks(n)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
# =====
# FIGURE 4 – FFT REAL + IMAGINARY TOGETHER
# =====
plt.figure(figsize=(11, 6))
plt.stem(
    n,
    fft_real,
    linefmt='C0-',
    markerfmt='C0o',
    basefmt=' ',
    label="Real Part"
)
plt.stem(
    n,
    fft_imag,
    linefmt='C1-',
    markerfmt='C1s',
    basefmt=' ',
    label="Imaginary Part"
)
# Real values
for i, value in enumerate(fft_real):

    offset = 10 if value >= 0 else -18

    plt.annotate(
        f"R={value:.2f}",
        (i, value),
```

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        xytext=(0, offset),
        textcoords="offset points",
        ha="center",
        fontsize=9
    )
# Imaginary values
for i, value in enumerate(fft_imag):
    offset = 24 if value >= 0 else -32
    plt.annotate(
        f"I={value:.2f}",
        (i, value),
        xytext=(0, offset),
        textcoords="offset points",
        ha="center",
        fontsize=9
    )
plt.axhline(0, linewidth=0.8)
plt.title(
    "FFT - Real and Imaginary Components",
    fontsize=15,
    fontweight="bold"
)
plt.xlabel("Frequency Index (k)")
plt.ylabel("Value")
plt.xticks(n)
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
# =====
# FIGURE 5 – MAGNITUDE SPECTRUM
# =====
plt.figure(figsize=(10, 5))
plt.stem(
    n,
    magnitude_spectrum,
    linefmt='C2- ',
    markerfmt='C2o',
    basefmt=' '

```

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)
for i, value in enumerate(magnitude_spectrum):
    plt.annotate(
        f"{value:.2f}",
        (i, value),
        xytext=(0, 8),
        textcoords="offset points",
        ha="center",
        fontsize=10
    )

plt.title(
    "FFT Magnitude Spectrum",
    fontsize=15,
    fontweight="bold"
)
plt.xlabel("Frequency Index (k)")
plt.ylabel("|X[k]|")
plt.xticks(n)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
# =====
# FIGURE 6 – PHASE SPECTRUM
# =====
plt.figure(figsize=(10, 5))
plt.stem(
    n,
    phase_spectrum,
    linefmt='C3-',
    markerfmt='C3o',
    basefmt=' '
)
for i, value in enumerate(phase_spectrum):
    offset = 10 if value >= 0 else -18
    plt.annotate(
        f"{value:.2f}",
        (i, value),
        xytext=(0, offset),

```

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        textcoords="offset points",
        ha="center",
        fontsize=10
    )
plt.axhline(0, linewidth=0.8)
plt.title(
    "FFT Phase Spectrum",
    fontsize=15,
    fontweight="bold"
)
plt.xlabel("Frequency Index (k)")
plt.ylabel("Phase (radians)")
plt.xticks(n)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
# =====
# FIGURE 7 – RECONSTRUCTED SIGNAL
# =====
plt.figure(figsize=(10, 5))
plt.stem(
    n,
    reconstructed_signal,
    linefmt='C4-',
    markerfmt='C4o',
    basefmt=' '
)
for i, value in enumerate(reconstructed_signal):
    plt.annotate(
        f"{value:.2f}",
        (i, value),
        xytext=(0, 8),
        textcoords="offset points",
        ha="center",
        fontsize=10
    )
plt.title(
    "Reconstructed Signal using IFFT",
    fontsize=15,

```

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```

        fontweight="bold"
    )
    plt.xlabel("Sample Index (n)")
    plt.ylabel("Amplitude")
    plt.xticks(n)
    plt.grid(True, alpha=0.3)
    plt.tight_layout()
    plt.show()

```

Output :-

```

PS D:\sem-5\dsip> python -u "d:\sem-5\dsip\EXP5.py"
=====
                        FFT ANALYSIS
=====

Original Signal:
[1 2 3 4 5 6 7 8]

FFT Result:
X[0] = 36.00 + j0.00
X[1] = -4.00 + j9.66
X[2] = -4.00 + j4.00
X[3] = -4.00 + j1.66
X[4] = -4.00 + j0.00
X[5] = -4.00 - j1.66
X[6] = -4.00 - j4.00
X[7] = -4.00 - j9.66

Magnitude Spectrum:
[36.   10.45  5.66  4.33  4.   4.33  5.66 10.45]

```

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x[n] = 1:8; % Discrete Time Signal

% Magnitude Spectrum
mag_spec = abs(fft(x))/length(x);

% Phase Spectrum
phase_spec = angle(fft(x));

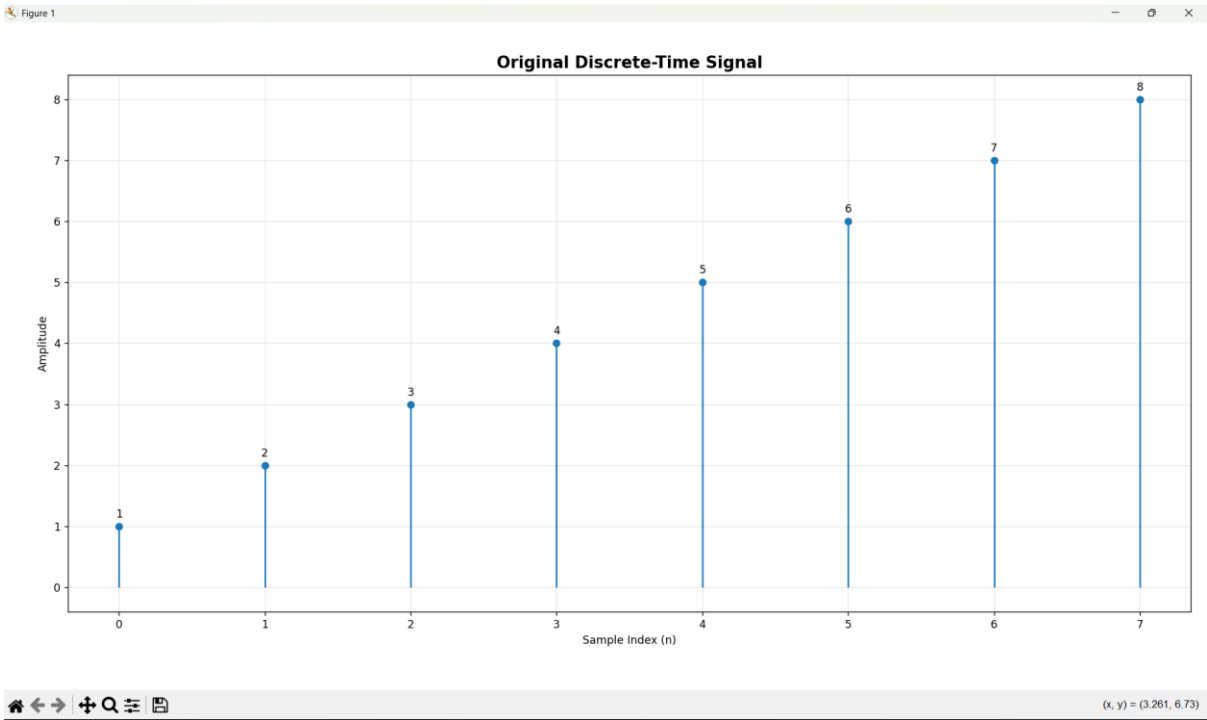
% Reconstructed Signal
reconstructed_signal = ifft(fft(x));

% Display Results
disp('Magnitude Spectrum:');
disp(mag_spec);

disp('Phase Spectrum:');
disp(phase_spec);

disp('Reconstructed Signal:');
disp(reconstructed_signal);

```





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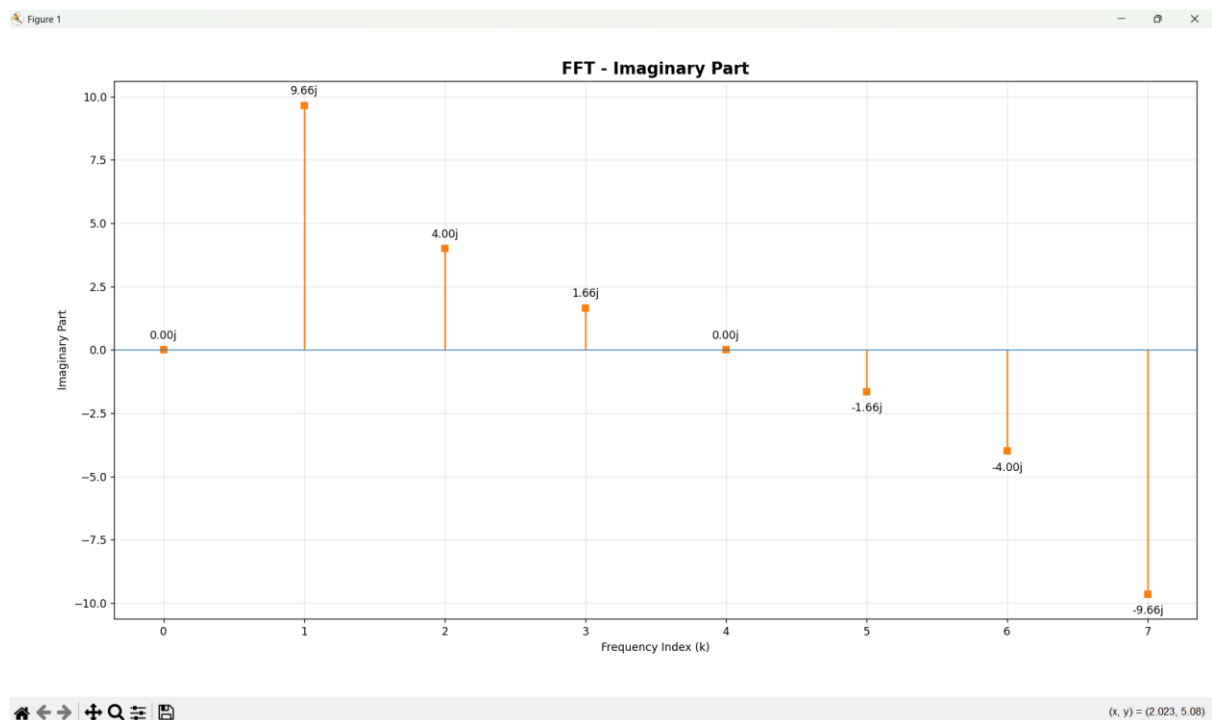
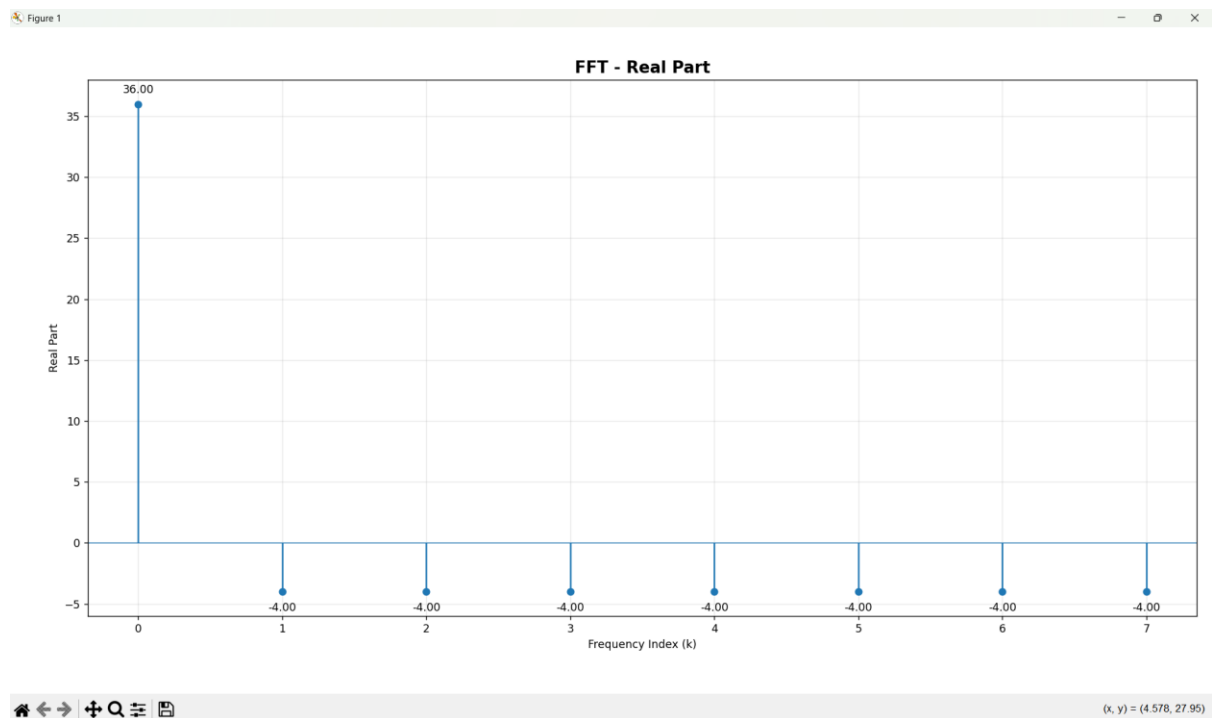
Subject Code: 01CT1513

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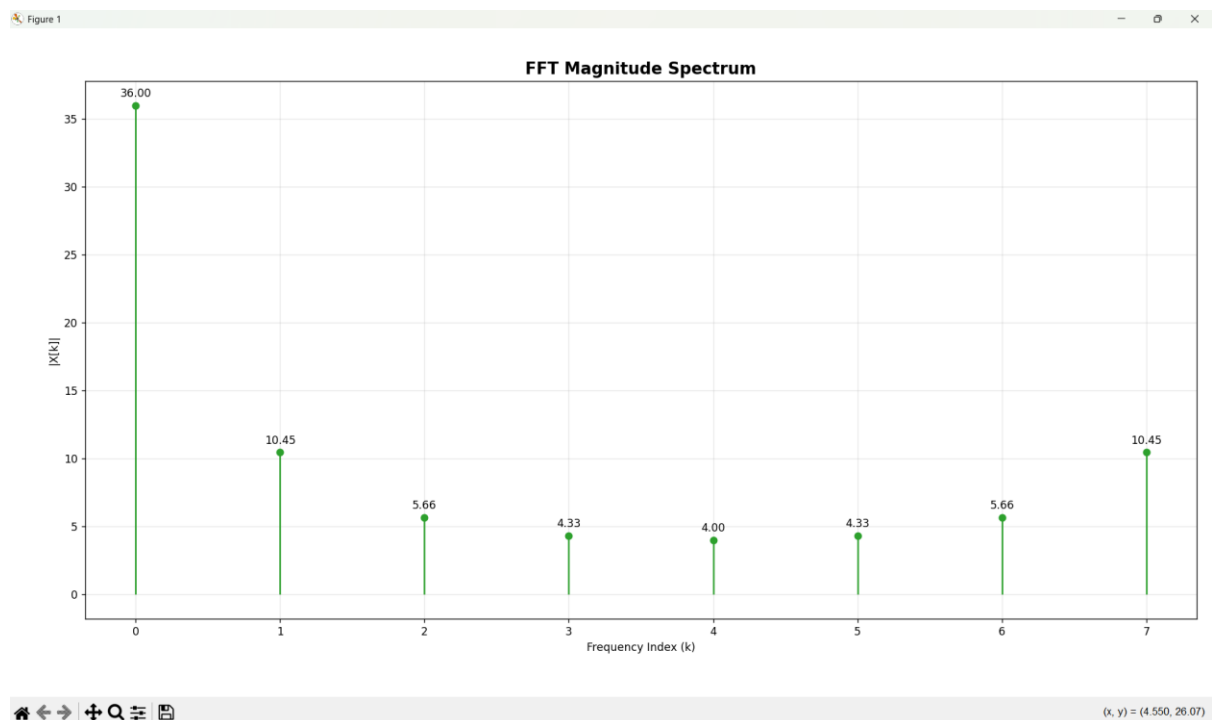
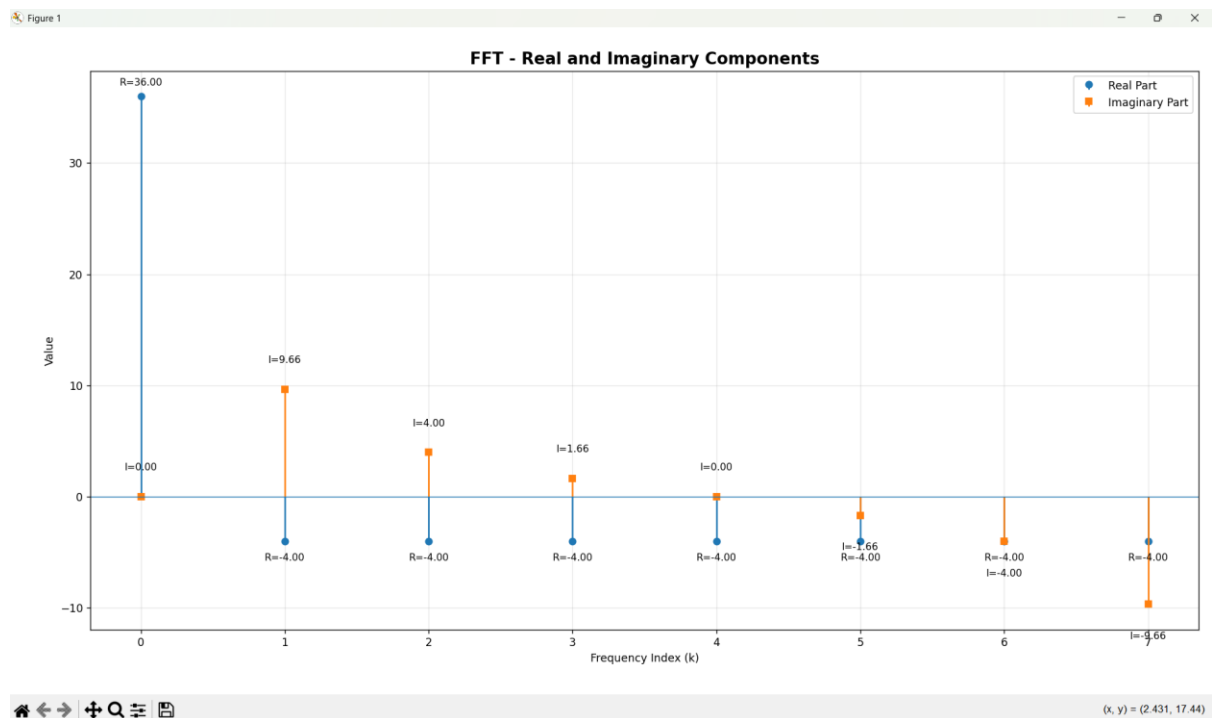
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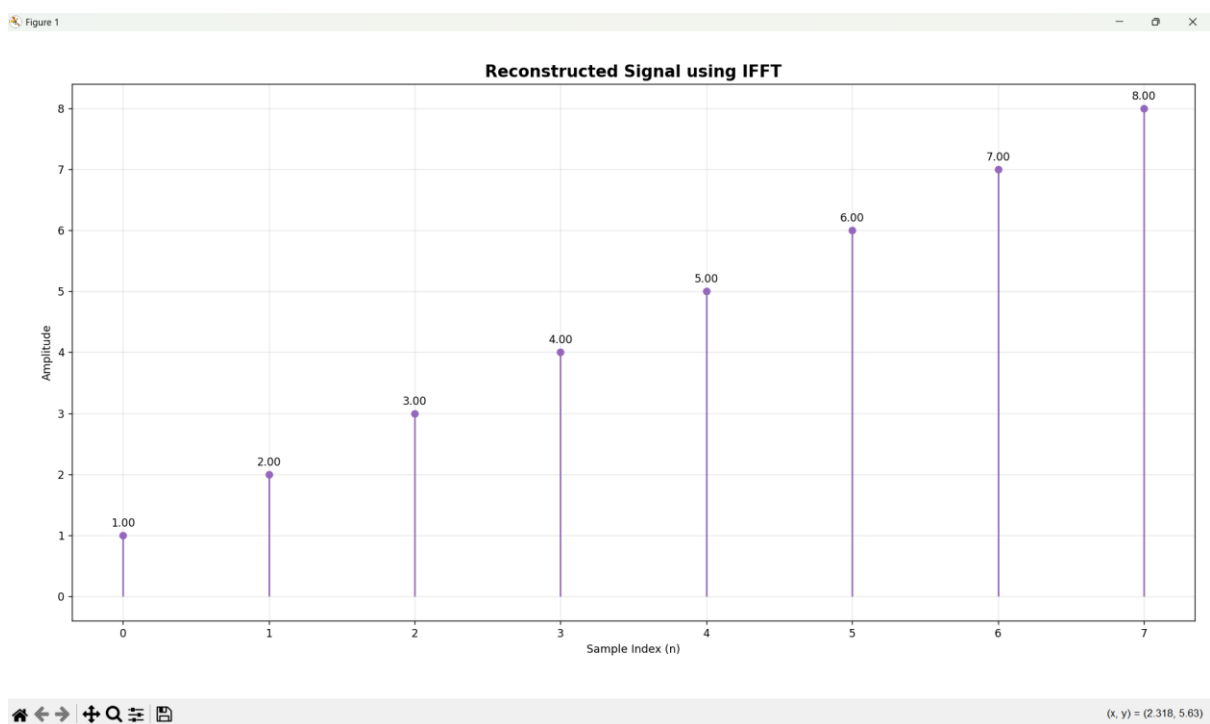
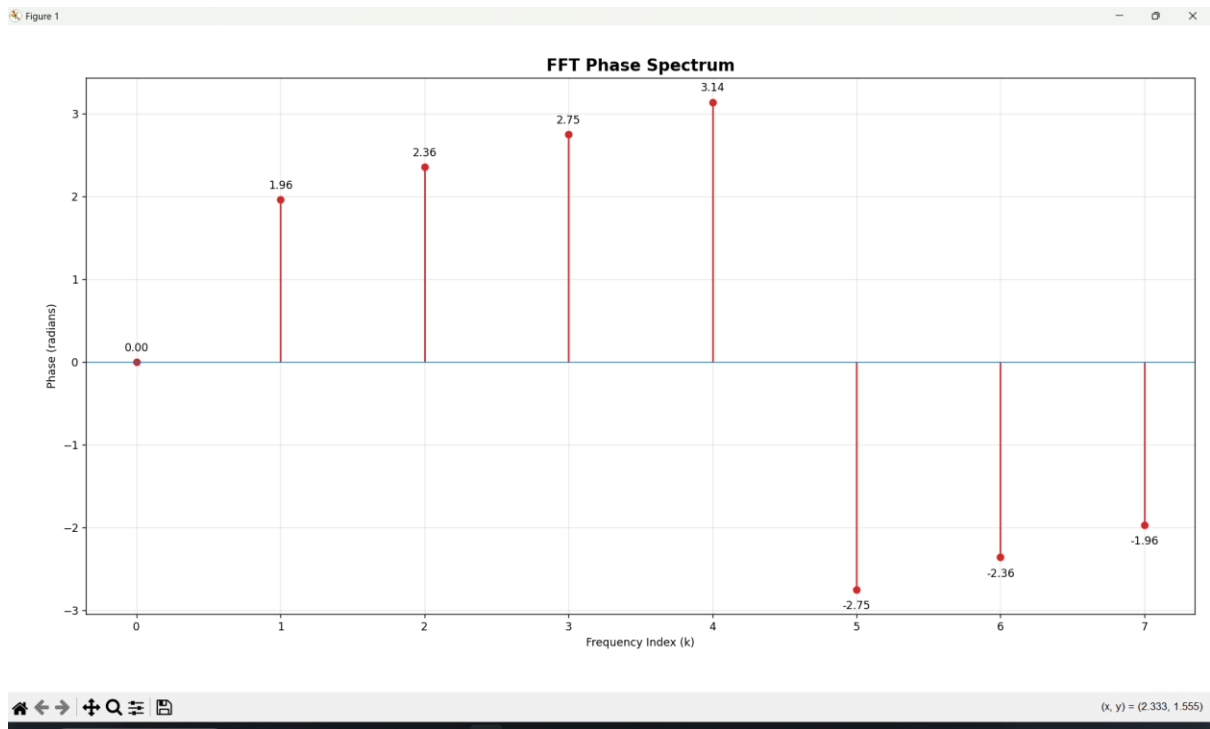
Subject Code: 01CT1513

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Conclusion

1. The given discrete-time signal was successfully analyzed using **FFT**.
2. FFT converted the signal from the **time domain to the frequency domain**.
3. The FFT result contains both **real and imaginary components**.
4. The magnitude spectrum shows the strength of each frequency component.
5. The phase spectrum provides the phase information of the frequency components.
6. The frequency-domain representation of the signal was successfully plotted.
7. The **real and imaginary parts** of the FFT were visualized separately and together.
8. The IFFT was applied to the FFT result to convert it back into the time domain.
9. The reconstructed signal was found to be almost identical to the original signal.
10. This verifies that FFT and IFFT preserve the information of the discrete-time signal.
11. The experiment demonstrates the usefulness of FFT for **frequency-domain signal analysis**.
12. Hence, FFT and IFFT were successfully performed on the given discrete-time signal.